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DOWNHOLE TOOL INCLUDING RELATED LUG SLOTS AND LUGS FOR COUPLING A MILLING TOOL AND WHIPSTOCK ASSEMBLY

Abstract

Provided is a milling tool, a whipstock assembly, a well system, and a method. The milling tool, in at least one aspect, includes a plurality of blades extending radially outward from a mill body, each of the plurality of blades having a primary surface and first and second side surfaces coupled to opposing sides of the primary surface, adjacent blades of the plurality of blades separated by a spacing. The milling tool may further includes a lug slot extending partially into one of the plurality of blades from the first side surface exposed by the spacing, the lug slot configured to engage with a lug of a related tool to couple the milling tool with the related tool when rotating in a first direction and to disengage from the lug of the related tool to release the milling tool from the related tool when rotating in a second opposite direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a divisional of U.S. patent application Ser. No. 18/333,079, entitled “DOWNHOLE TOOL INCLUDING RELATED LUG SLOTS AND LUGS FOR COUPLING A MILLING TOOL AND WHIPSTOCK ASSEMBLY”, filed on Jun. 12, 2023. The above-listed application is commonly assigned with the present application is incorporated herein by reference as if reproduced herein in its entirety.

BACKGROUND

[0002] The unconventional market is very competitive. The market is trending towards longer horizontal wells to increase reservoir contact. Multilateral wells offer an alternative approach to maximize reservoir contact. Multilateral wells include one or more lateral wellbores (e.g., secondary wellbores) extending from a main wellbore (e.g., primary wellbore). A lateral wellbore is a wellbore that is diverted from the main wellbore or another lateral wellbore.

[0003] Lateral wellbores are typically formed by positioning one or more deflector assemblies (e.g., whipstock assemblies) at desired locations in the main wellbore (e.g., an open hole section or cased hole section) with a running tool. The deflector assemblies are often laterally and rotationally fixed within the primary wellbore using an anchoring assembly.

Description

BRIEF DESCRIPTION

[0004] Reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0005] FIG. 1 illustrates a schematic view of a well system designed, manufactured and operated according to one or more embodiments disclosed herein;

[0006] FIGS. 2A through 2E illustrate various different views of a two part milling and running tool designed, manufactured and operated according to one or more embodiments of the disclosure;

[0007] FIGS. 3A and 3B illustrate various different views of a milling tool designed, manufactured and/or operated according to one or more embodiments of the disclosure;

[0008] FIGS. 4A and 4B illustrate various different views of a milling tool designed, manufactured and/or operated according to one or more alternative embodiments of the disclosure;

[0009] FIGS. 5A and 5B illustrate various different views of a whipstock assembly designed, manufactured and/or operated according to one or more embodiments of the disclosure; and

[0010] FIGS. 6 through 13C illustrate one embodiment for designing, manufacturing and/or operating a downhole tool 600 according to the present disclosure.

DETAILED DESCRIPTION

[0011] In the drawings and descriptions that follow, like parts are typically marked throughout the specification and drawings with the same reference numerals, respectively. The drawn figures are not necessarily to scale. Certain features of the disclosure may be shown exaggerated in scale or in somewhat schematic form and some details of certain elements may not be shown in the interest of clarity and conciseness. The present disclosure may be implemented in embodiments of different forms.

[0012] Specific embodiments are described in detail and are shown in the drawings, with the understanding that the present disclosure is to be considered an exemplification of the principles of the disclosure, and is not intended to limit the disclosure to that illustrated and described herein. It

is to be fully recognized that the different teachings of the embodiments discussed herein may be employed separately or in any suitable combination to produce desired results.

[0013] Unless otherwise specified, use of the terms “connect,” “engage,” “couple,” “attach,” or any other like term describing an interaction between elements is not meant to limit the interaction to a direct interaction between the elements and may also include an indirect interaction between the elements described. Unless otherwise specified, use of the terms “up,” “upper,” “upward,” “uphole,” “upstream,” or other like terms shall be construed as generally away from the bottom, terminal end of a well; likewise, use of the terms “down,” “lower,” “downward,” “downhole,” “downstream,” or other like terms shall be construed as generally toward the bottom, terminal end of a well, regardless of the wellbore orientation. Use of any one or more of the foregoing terms shall not be construed as denoting positions along a perfectly vertical axis. Unless otherwise specified, use of the term “subterranean formation” shall be construed as encompassing both areas below exposed earth and areas below earth covered by water, such as ocean or fresh water.

[0014] The disclosure addresses the challenge of running a whipstock assembly on a mill, for example in an effort to reduce trip count. With this in mind, in one embodiment the present disclosure provides a two part drilling and running tool (e.g., including a smaller assembly and a larger bit assembly) that may be used to run a whipstock assembly downhole. In at least one embodiment, the larger bit assembly (e.g., uphole/larger bit assembly) is connected to the whipstock assembly, for example proximate an uphole end of the whipstock assembly. At a desired point in time, the smaller assembly may pull back uphole and connect to the larger bit assembly, thereby forming a new combined bit assembly (e.g., that looks and functions like a conventional lead mill). For purposes of the present disclosure, the term bit assembly is intended to encompass both mill assemblies and drill bit assemblies. Following the successful creation of the exit and the drilling of the lateral, the lateral completion could be installed and then tied together with the main bore by installing a junction (e.g., level 5 junction in one embodiment).

[0015] In accordance with at least one embodiment of the disclosure, the smaller assembly is a smaller bit assembly having one or more cutting features (e.g., teeth, blades, etc.) thereon. The smaller assembly, in one embodiment, could be connected to a conveyance that extends through the larger bit assembly and is then connected to the rest of the drill string, or perhaps to a downhole motor directly. In at least one embodiment, the smaller assembly is sized such that it can wholly or partially fit into a bore of the whipstock assembly, such that in one embodiment it may connect to the whipstock assembly or there below. In at least one embodiment, the larger bit assembly and/or smaller assembly are coupled to the whipstock assembly or other downhole device there below using a shear feature, a collection of lugs and/or slots or other connecting mechanism. With the smaller assembly free from the whipstock assembly or other downhole device there below, the smaller assembly is free to slide back uphole and into the larger bit assembly.

[0016] Once the smaller assembly has been fully retracted into the larger bit assembly it can be secured to the larger bit assembly for the milling operation. In at least one embodiment, a simple snap ring falls into a groove in the smaller assembly, thereby securing (e.g., laterally securing) the smaller assembly within the larger bit assembly. Many alternate methods are possible, such as spring-loaded pins, a thread, or an interference fit between the two bit assemblies.

[0017] At this point the larger bit assembly may be disconnected from the whipstock assembly tip and a normal window can be milled in the casing and/or formation as is current industry practice. As is sometimes the practice with milling windows, secondary mills (e.g., watermelon mills) may be added to follow the lead mill to ensure proper window geometry. Likewise multiple trips may be required to successfully mill a window. In those cases, extra mills or trips could be performed as is done today. Thereafter, the remainder of the multilateral construction may be completed, for example including placing a multilateral junction including a mainbore leg and a lateral bore leg at the junction between the main wellbore and the lateral wellbore.

[0018] Up to this point, the use of a two part drilling and running tool has been discussed for

creating an exit window from a cased mainbore. An alternate use for this new technology is to sidetrack from an open-hole main bore. In this alternate use, the bit assembly would be more appropriately called a drill bit, as it would be drilling formation to exit the main bore rather milling casing. This would be useful for simple sidetracking where the main bore may need to be abandoned, or it may be used during the construction of an open-hole multilateral junction. In this use, the smaller assembly and larger bit assembly would be designed differently than what is shown herein, for example to closely resemble a drill bit instead of a mill bit. This would necessitate certain changes to the external cutting features, which should be understood to not deviate from the core features described herein.

[0019] FIG. 1 is a schematic view of a well system **100** designed, manufactured and operated according to one or more embodiments disclosed herein. The well system **100** includes a platform **120** positioned over a subterranean formation **110** located below the earth's surface **115**. The platform **120**, in at least one embodiment, has a hoisting apparatus **125** and a derrick **130** for raising and lowering one or more downhole tools including one or more conveyance (e.g., pipe strings, such as a drill string **140**). Although a land-based oil and gas platform **120** is illustrated in FIG. 1, the scope of this disclosure is not thereby limited, and thus could potentially apply to offshore applications. The teachings of this disclosure may also be applied to other land-based and/or water-based well systems different from that illustrated.

[0020] As shown, a main wellbore **150** has been drilled through the various earth strata, including the subterranean formation **110**. The term “main” wellbore is used herein to designate a primary wellbore from which another secondary wellbore is drilled. It is to be noted, however, that a main wellbore **150** does not necessarily extend directly to the earth's surface **115**, but could instead be a branch of yet another lateral wellbore. A casing string **160** may be at least partially cemented within the main wellbore **150**. The term “casing” is used herein to designate a tubular string used to line a wellbore. Casing may actually be of the type known to those skilled in the art as a “liner” and may be made of any material, such as steel or composite material and may be segmented or continuous, such as coiled tubing. The term “lateral” wellbore is used herein to designate a wellbore that is drilled outwardly from its intersection with another wellbore, such as a main wellbore. Moreover, a lateral wellbore may have another lateral wellbore drilled outwardly therefrom.

[0021] A whipstock assembly **170** according to one or more embodiments of the present disclosure may be positioned at a location in the main wellbore **150**. Specifically, the whipstock assembly **170** could be placed at a location in the main wellbore **150** where it is desirable for a lateral wellbore **180** to exit. Accordingly, the whipstock assembly **170** may be used to support a drilling/milling tool used to penetrate a window in the main wellbore **150**. In at least one embodiment, once the window has been milled and a lateral wellbore **180** formed, the whipstock assembly **170** may be retrieved and returned uphole by a retrieval tool, in some embodiments in only a single trip.

[0022] In some embodiments, an anchoring assembly **190** may be placed downhole in the wellbore **150** to support and anchor downhole tools, such as the whipstock assembly **170**, for keeping the whipstock assembly **170** in place while milling the casing **160** and/or drilling the lateral wellbore **180**. The anchoring assembly **190**, in accordance with the disclosure, may be employed in a cased section of the main wellbore **150**, or may be located in an open-hole section of the main wellbore **150**, as is shown. As such, the anchoring assembly **190** in at least one embodiment may be configured to resist at least 6,750 newton meters (Nm) (e.g., about 5,000 lb-ft) of torque. In yet another embodiment, the anchoring assembly **190** may be configured to resist at least 13,500 newton meters (Nm) (e.g., about 10,000 lb-ft) of torque, and in yet another embodiment configured to resist at least 20,250 newton meters (Nm) (e.g., about 15,000 lb-ft) of torque. Similarly, the anchoring assembly **190** may be configured to resist at least 1814 kg (e.g., about 4,000 lb) of axial force. In yet another embodiment, the anchoring assembly **190** may be configured to resist at least 4536 kg (e.g., about 10,000 lb) of axial force, and in yet another embodiment the anchoring assembly **190** may be configured to resist at least 6804 kg (e.g., about 15,000 lb) of axial force.

[0023] In the illustrated embodiment, the anchoring assembly **190** may be a hydraulically activated anchoring assembly. In this embodiment, once the anchoring assembly **190** reaches a desired location in the main wellbore **150**, fluid pressure may be applied to set the hydraulic anchoring assembly. In at least one embodiment, the hydraulically activated anchoring assembly includes two or more hydraulic activation chambers, and the activation fluid is supplied to the two or more hydraulic activation chambers (e.g., through a two-part milling assembly coupled to the whipstock assembly **170**) to move the two or more hydraulic activation chambers from the first collapsed state to the second activated state and engage a wall of the main wellbore **150**. The anchoring assembly **190** may also include, in some embodiments, an expandable medium positioned radially about the two or more hydraulic activation chambers. In some aspects, the expandable medium may be configured to grip and engage the wall of the main wellbore **150** when the two or more hydraulic activation chambers are in the second activated state. Notwithstanding, other fluid activated anchoring assemblies (e.g., other than those having two or more hydraulic activation chambers) may be used and remain within the scope of the disclosure. In at least one other embodiment, the hydraulically activated anchoring assembly includes one or more hydraulic activation slips, and the activation fluid is supplied to the one or more hydraulic activation slips (e.g., through a two-part milling assembly coupled to the whipstock assembly **170**) to move the one or more hydraulic activation slips from the first collapsed state to the second activated state and engage the wall of the main wellbore **150**.

[0024] Furthermore, mechanical activated anchoring assemblies could also be used and remain within the scope of the disclosure. For instance, in yet other embodiments, the anchoring assembly **190** is a latch coupling. In this embodiment, the latch coupling (e.g., a profile in the casing engages with a reciprocal profile in the whipstock assembly **170**) anchors the whipstock assembly **170**, and any other features hanging there below (e.g., screens, valves, etc.) in the casing string **160**. Once the anchoring assembly **190** reaches a desired location in the main wellbore **150**, the reciprocal profile in the whipstock assembly **170** may be activated to engage with the latch coupling profile in the casing string **160**, thereby setting the anchoring assembly **190**. Thus, in at least one embodiment, the anchoring assembly **190** is not hydraulically activated, but is mechanically activated.

[0025] In at least one embodiment, a multilateral junction is positioned at an intersection between the resulting main wellbore **150** and the resulting lateral wellbore **180**. In accordance with one embodiment, the multilateral junction might include a main bore leg forming a first pressure tight seal with the main bore completion and a lateral bore leg forming a second pressure tight seal with the lateral bore completion, such that the main bore completion and the lateral bore completion are hydraulically isolated from one another. What results, in one or more embodiments, is an open hole TAML Level 5 pressure tight junction.

[0026] Turning now to FIGS. 2A through 2E, illustrated are various different views of a two part milling and running tool **200** designed, manufactured and operated according to one or more embodiments of the disclosure. FIG. 2A illustrates a side view of the two part milling and running tool **200**. FIG. 2B illustrates an enlarged side view of a larger bit assembly of FIG. 2A. FIG. 2C illustrates an isometric view of one embodiment of an internal profile of the larger bit assembly of FIG. 2A. FIG. 2D illustrates an enlarged side view of a smaller assembly of FIG. 2A. FIG. 2E illustrates an isometric view of one embodiment of the smaller assembly of FIG. 2A.

[0027] With initial reference to FIG. 2A, the two part milling and running tool **200**, in the illustrated embodiment, includes a conveyance **210**. The conveyance **210**, in at least one embodiment, is a tubular, such as jointed pipe or coiled tubing. The two part milling and running tool **200**, as shown in the embodiment of FIGS. 2A through 2E, may additionally include a larger bit assembly **220** and a smaller assembly **250** coupled thereto. As indicated above, the phrase “bit assembly,” as used herein, is intended to include both milling assemblies (e.g., as might be used to mill through casing) and drill bit assemblies (e.g., as might be used to drill through formation), as

well as any combination of the two. As discussed above, and shown in the FIGs., the smaller assembly **250** may also be a smaller bit assembly, and thus may contain one or more different types of cutting features along a downhole face thereof.

[0028] In the illustrated embodiment of FIGS. 2A through 2E, the smaller assembly **250** is coupled to an end (e.g., coupled proximate, such as within 2 meters, if not within 1 meter, if not within 0.5 meters, if not within 0.1 meters, a downhole end) of the conveyance **210**, whereas the larger bit assembly **220** is in sliding engagement with the conveyance **210**. Accordingly, assuming that something (e.g., friction, a shear feature, etc.) is holding the larger bit assembly **220** in place as the conveyance **210** is moved, the smaller assembly **250** may slide relative to the larger bit assembly **220**. For instance, if the conveyance **210** were withdrawn uphole, the larger bit assembly **220** would slide along the conveyance **210**, thereby allowing the smaller assembly **250** to slide toward the larger bit assembly **220**.

[0029] As will be discussed in greater detail below, the two part milling and running tool **200** may be used to deploy a whipstock assembly, and thus may be coupled to the whipstock assembly when running downhole. In at least one embodiment, the larger bit assembly **220** is coupled to the whipstock assembly, for example using a collection of lug slot(s) and lug(s). The smaller assembly **250** may also be coupled to one of the whipstock assembly, the anchoring assembly or the wellbore liner, in at least one embodiment, which would prevent the smaller assembly **250** from sliding toward the larger bit assembly **220** during the run-in-hole phase. Only when the smaller assembly **250** is decoupled from the whipstock assembly, the anchoring assembly or the wellbore liner, would the smaller assembly **250** be allowed to slide toward the larger bit assembly **220** (e.g., assuming that the larger bit assembly **220** is still coupled to the whipstock assembly or held in place with friction).

[0030] In the illustrated embodiment of FIG. 2A, the two part milling and running tool **200** is positioned in the run-in-hole position. In this run-in-hole position, the larger bit assembly **220** would be spaced apart from the smaller assembly **250** by a distance (D.sub.0). In at least one embodiment, the distance (D.sub.0) approximates the length of the whipstock assembly that the two part milling and running tool **200** is coupled to. According to this embodiment, the smaller assembly **250** might couple proximate a downhole end of the whipstock assembly, whereas the larger bit assembly **220** might couple proximate an uphole end of the whipstock assembly. Thus, in at least one embodiment, the distance (D.sub.0) is at least 1 meter, or at least 2 meters. In yet another embodiment, the distance (D.sub.0) is at least 4 meters, and in even another embodiment the distance (D.sub.0) is at least 5 meters. In at least one other embodiment, the distance (D.sub.0) approximates the length of the whipstock assembly and at least a portion of the length of the anchoring assembly. In at least yet another embodiment, the distance (D.sub.0) approximates the length of the whipstock assembly, the length of the anchoring assembly, and at least a portion of the length of a wellbore liner.

[0031] In at least one embodiment, as discussed in greater detail below, the larger bit assembly **220** includes a mill body **222** having a plurality of blades **230** extending radially outward therefrom (e.g., whether extending directly radially outward therefrom or extending indirectly radially outward therefrom), each of the plurality of blades **230** having a primary surface and first and second side surfaces coupled to opposing sides of the primary surface. In at least one embodiment, adjacent blades of the plurality of blades are separated by a spacing **225**. Further to this embodiment, the larger bit assembly **220** may include a lug slot **235** extending partially into one of the plurality of blades **230** from the first side surface exposed by the spacing **225**. In one or more embodiments, the lug slot **235** is configured to engage with a lug (not shown) of a related tool (e.g., whipstock assembly) to couple the two part milling and running tool **200** with the related tool when rotating in a first direction, and to disengage from the lug (not shown) of the related tool to release the two part milling and running tool **200** from the related tool when rotating in a second opposite direction.

[0032] Turning now to FIG. 2B, the larger bit assembly **220** may have one or more cutting features **240** on the one or more blades **230**. While specific blades **230** and cutting features **240** are illustrated in FIG. 2B, any currently known or hereafter discovered blades and cutting features may be used and remain within the scope of the disclosure. The larger bit assembly **220**, in the illustrated embodiment, includes a cutting diameter (d.sub.l). In at least one embodiment, the cutting diameter (d.sub.l) approximates the size of an opening (e.g., in the casing and/or formation) forming a lateral wellbore.

[0033] Turning now to FIG. 2C, in at least one embodiment, the larger bit assembly **220** may also include a lock ring profile **245**, which may be configured to hold a lock ring (not shown) that could ultimately engage with an associated lock ring profile in the smaller assembly **250**, or vice versa.

[0034] Turning now to FIG. 2D, the smaller assembly **250** may have one or more blades **260** and one or more cutting features **265** thereon, thereby making the smaller assembly **250** a smaller bit assembly. While specific blades **260** and cutting features **265** are illustrated in FIG. 2D, any currently known or hereafter discovered blades and cutting features may be used and remain within the scope of the disclosure. The smaller assembly **250**, in the illustrated embodiment, includes a cutting diameter (d.sub.s). In at least one embodiment, the cutting diameter (d.sub.s) is at least 10 percent less than the cutting diameter (d.sub.l). In at least one embodiment, the cutting diameter (d.sub.s) is at least 25 percent less than the cutting diameter (d.sub.l), in yet another embodiment at least 50 percent less than the cutting diameter (d.sub.l), in yet another embodiment at least 75 percent less than the cutting diameter (d.sub.l), and in yet another embodiment at least 90 percent less than the cutting diameter (d.sub.l). The smaller assembly **250**, in the illustrated embodiment, may further include an associated lock ring profile **270**. Accordingly, the lock ring profile **270**, as well as the associated lock ring profile **245** of the larger bit assembly **220**, and a lock ring (not shown), may be used to linearly fix the larger bit assembly **220** and the smaller assembly **250**.

[0035] Turning now to FIG. 2E, the smaller assembly **250** may additionally include one or more fluid ports **275**. The one or more fluid ports **275**, in the illustrated embodiment, may provide fluid access past the smaller assembly **250**, to help cool the bit/mill, lubricate and remove cuttings. In yet another embodiment, the one or more fluid ports **275**, provide fluid access past the smaller assembly **250**, particularly, when the smaller assembly **250** is coupled to and sealed with the whipstock assembly. For example, the one or more fluid ports **275** may be fluidly coupled with a through bore in the whipstock assembly, and thus may be used to activate a hydraulic wellbore anchoring assembly, among other downhole features.

[0036] Turning to FIGS. 3A and 3B, illustrated are various different views of a milling tool **300** designed, manufactured and/or operated according to one or more embodiments of the disclosure. The milling tool **300**, in the illustrated embodiment, includes a mill body **310**. In at least one embodiment, the mill body **310** is a tubular. The milling tool **300**, in one or more embodiments, may additionally include a plurality of blades **330** extending radially outward therefrom, each of the plurality of blades **330** having a primary surface **331** and first and second side surfaces **332**, **333** coupled to opposing sides of the primary surface **331**. In the illustrated embodiment, adjacent blades of the plurality of blades **330** are separated by a spacing **325**.

[0037] Further to the embodiment of FIGS. 3A and 3B, the milling tool **300** may additionally include a lug slot **335** extending partially into one of the plurality of blades **330** from the first side surface **332** exposed by the spacing **325**. In accordance with this embodiment, the lug slot **335** is configured to engage with a lug (not shown) of a related tool (e.g., whipstock assembly) to couple the milling tool **300** with the related tool. For example, the lug slot **335** may engage the milling tool **300** with the related tool when rotating in a first direction, and disengage from the lug of the related tool to release the milling tool from the related tool when rotating in a second opposite direction. In the illustrated embodiment, the lug slot **335** has a larger lug slot portion **336** radially proximate the mill body **310** and a smaller lug slot portion **337** radially distal the mill body **310**. For example, in one or more embodiments, the lug slot **335** is an inverted T-shaped lug slot, as shown.

[0038] Further to the embodiment of FIGS. 3A and 3B, the lug slot **335** is a first lug slot **335a**, and the milling tool **300** further includes a second lug slot **335b** extending partially into one of the plurality of blades **330** from a first side surface **332** exposed by a spacing **325**. In the illustrated embodiment, the first lug slot **335a** and the second lug slot **335b** are located in a same one of the plurality of blades **330**. In another embodiment, the first lug slot **335a** is located in a first of the plurality of blades **330** and the second lug slot **335b** is located in a second of the plurality of blades **330**. While only first and second lug slots **335a**, **335b** are illustrated in the embodiment of FIGS. 3A and 3B, other embodiments exist wherein more than two lug slots **335** are located in a single one of the plurality of blades **330**, or more than two lug slots **335** are located in two or more of the plurality of blades **330**.

[0039] In at least one embodiment, the lug slot(s) **335** are oriented such that any axial loading (e.g., uphole and downhole axial loading) will communicate between the mill body **310** (e.g., high strength steel mill body) and the lug in the related device (e.g., lug in the whipstock assembly). In at least one embodiment, the lug slot(s) **335** are also oriented so that right hand rotational communication through the conveyance (e.g., drill string) will communicate rotational loading between the mill body **310** and the lug in the related device (e.g., lug in the whipstock assembly). Similarly, left hand rotation will communicate rotational loading between the lug in the related device (e.g., lug in the whipstock assembly) and a finger portion in a related shifting sleeve, which is acceptable since there is generally no need to communicate high levels of left hand torque.

[0040] In the illustrated embodiment of FIGS. 3A and 3B, the milling tool **300** may have one or more cutting features **340** on the one or more blades **330**. While specific blades **330** and cutting features **340** are illustrated in FIGS. 3A and 3B, any currently known or hereafter discovered blades and cutting features may be used and remain within the scope of the disclosure. Further to the embodiment of FIGS. 3A and 3B, the milling tool **300** may have a seal groove **350** (e.g., as might include a seal member) therein.

[0041] Turning to FIGS. 4A and 4B, illustrated are various different views of a milling tool **400** designed, manufactured and/or operated according to one or more alternative embodiments of the disclosure. The milling tool **400** of FIGS. 4A and 4B is similar in many respects to the milling tool **300** of FIGS. 3A and 3B. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The milling tool **400** differs, for the most part, from the milling tool **300**, in that the milling tool **400** additionally includes a sliding sleeve **410** positioned about the mill body **310** proximate the plurality of blades **330**. In at least one embodiment, the sliding sleeve **410** includes a finger portion **420**, the sliding sleeve **410** configured to move between a first position (e.g., that shown in FIGS. 4A and 4B) locating the finger portion **420** within the spacing **325** proximate the lug slot **335**, and a second position (e.g., not shown) removing the finger portion **420** from the spacing **325** proximate the lug slot **335**. In at least one embodiment, the first position is configured to keep the lug of the related tool engaged within the lug slot **335**, and the second position is configured to allow the lug of the related tool to disengage from the lug slot **335**.

[0042] In one or more embodiments, the plurality of blades **330** are angled relative to an axis of rotation **430** of the mill body **310**, and furthermore the finger portion **420** is angled relative to the axis of rotation **430** of the mill body **310**. Accordingly, in at least one embodiment, a first angle ($\theta_{\text{sub.1}}$) of the plurality of blades **330** is substantially identical to a second angle ($\theta_{\text{sub.2}}$) of the finger portion **420**. The term “substantially identical,” as used herein with respect to these angles, means that the angles are within 15 percent of each other. In yet another embodiment, the first angle ($\theta_{\text{sub.1}}$) of the plurality of blades **330** is ideally identical to the second angle ($\theta_{\text{sub.2}}$) of the finger portion **420**, wherein “ideally identical” means that the angles are within 10 percent of each other. In yet even another embodiment, the first angle ($\theta_{\text{sub.1}}$) of the plurality of blades **330** is exactly identical to the second angle ($\theta_{\text{sub.2}}$) of the finger portion **420**, wherein “exactly identical” means that the angles are within 5 percent of each other. In yet another embodiment, the first angle ($\theta_{\text{sub.1}}$) of the plurality of blades **330** is perfectly identical to the second angle ($\theta_{\text{sub.2}}$) of the

finger portion **420**, wherein “perfectly identical” means that the angles are within 2 percent of each other. In another embodiment, the first angle ($\theta_{sub.1}$) and the second angle ($\theta_{sub.2}$) are the same. In yet another embodiment, the plurality of blades **330** are curved and the finger portion **420** is curved, for example using the same similarities (e.g., identical values) as to when they are angled. [0043] As will be further understood below, the sliding sleeve **410** may be a hydraulically activated sliding sleeve, an increase in pressure within the mill body **310** configured to move the sliding sleeve from the first position to the second position. In yet another embodiment, the sliding sleeve **410** is a mechanically activated sliding sleeve, an external mechanical movement configured to move the sliding sleeve **410** from the first position to the second position. In an effort to keep the sliding sleeve **410** in the first position while running downhole, a shear feature (e.g., shear pin, shear screw, etc.) may exist between the sliding sleeve **410** and the mill body **310**.

[0044] The milling tool **400**, in one or more other embodiments, may include a secondary mill **440**. For example, in one or more embodiments, the milling tool **400** could include a watermelon mill as the secondary mill **440**. Those skilled in the art understand the benefits of including the secondary mill **440** with the milling tool **400**.

[0045] Turning now to FIGS. 5A and 5B, illustrated is one embodiment of a whipstock assembly **500** designed, manufactured and/or operated according to one or more embodiments of the disclosure. The whipstock assembly **500**, in one embodiment, includes a whipstock body **510**. The whipstock body **510**, in at least one embodiment, has an uphole end **520** and a downhole end **525**. Further to the embodiment of FIGS. 5A and 5B, the whipstock body **510** may define a taperface **530a** and may have a concave portion **530b**.

[0046] In accordance with one embodiment of the disclosure, the whipstock assembly **500** may additionally include a lug **535** coupled proximate the uphole end **520** of the whipstock body **510**. The term “proximate,” as used with respect to the lug **535**, means within an uphole 30 percent of the whipstock body. In yet another embodiment, the lug **535** is coupled substantially proximate the uphole end **520** of the whipstock body **510**, wherein the term “substantially proximate” means within an uphole 20 percent of the whipstock body. In yet another embodiment, the lug **535** is coupled ideally proximate the uphole end **520** of the whipstock body **510**, wherein the term “ideally proximate” means within an uphole 10 percent of the whipstock body.

[0047] In at least one embodiment, the lug **535** may be coupled to the concave portion **530b** of the whipstock body **510**. In accordance with one embodiment, the lug **535** is configured to engage with a lug slot (e.g., not shown, but in one embodiment the lug slot **335** of FIGS. 3A through 4B) of a related milling tool (e.g., not shown, but in one embodiment the milling tool **300**, **400** of FIGS. 3A through 4B) to couple the whipstock assembly **500** with the related milling tool. For example, the lug **535** may engage with the related lug slot when the milling tool rotates in a first direction, and to disengage from the related lug slot of the related milling tool to release the whipstock assembly **500** from the related milling tool when the milling tool rotates in a second opposite direction.

[0048] In one or more embodiments, such as that shown, the lug **535** has a smaller lug portion **536** and a larger lug portion **537**. Further to this embodiment, the smaller lug portion **536** may be more proximate the whipstock body **510** than the larger lug portion **537**. In accordance with one or more embodiments, the lug **535** is a T-shaped lug, the smaller lug portion **536** coupled to the whipstock body **510**. In at least one other embodiment, a first footprint of a portion of the lug **535** proximate the whipstock body **510** is different than a second footprint of a portion of the lug **535** distal the whipstock body **510**. For example, in one or more embodiments the first footprint is smaller than the second footprint.

[0049] In the illustrated embodiment, the lug **535** is a first lug **535a**, and the whipstock assembly **500** further includes a second lug **535b** coupled proximate the uphole end **520** of the whipstock body **510**. In at least one embodiment, a centerline **540** taken through the first lug **535a** and the second lug **535b** is angled relative to a longitudinal centerline **545** of the whipstock body **510**. In yet another embodiment, an angle ($\theta_{sub.3}$) between the centerline **540** and the longitudinal

centerline **545** is substantially identical to an angle (θ .sub.1) of a plurality of blades of the related milling tool that the whipstock assembly is configured to engage (e.g., angle (θ .sub.1) of FIG. **4B**). [0050] The lug **535** may be formed in a number of different methods. In at least one embodiment, the lug **535** is integrally formed with the whipstock body **510**. In yet another embodiment, the lug **535** is coupled to the whipstock body **510** via an opening (not shown) in the whipstock body **510**. In yet another embodiment, a fastener (not shown) extends at least partially into the opening to couple the lug **535** to the whipstock body **510**. In even another embodiment, the log **535** is welded to the whipstock body **510**.

[0051] With reference to FIGS. **3A** through **5B**, the present disclosure creates a high strength connection between a milling tool **300**, **400** and a whipstock assembly **500**. Traditionally, this connection is a shear feature with maximum axial rating of less than 100 kip, and maximum torque rating of less than 25,000 ft-lb. This new high strength connection, however, can achieve axial loads much greater than achievable with the current shear feature, since (e.g., among other reasons) the number, size and/or shape of the whipstock lugs can vary. Furthermore, the lug slots of the present disclosure may capture the lug of the present disclosure on all but one side.

[0052] Turning to FIGS. **6** through **13C**, illustrated is one embodiment for designing, manufacturing and/or operating a downhole tool **600** according to the present disclosure. The downhole tool **600** of FIGS. **6** through **13C** includes many of the same feature as the milling tool **400** of FIGS. **4A** and **4B** and whipstock assembly **500** of FIGS. **5A** and **5B**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features.

[0053] Turning to FIG. **6**, illustrated is a side view of the downhole tool **600** according to one embodiment of the disclosure. The downhole tool **600**, in this view, includes a milling tool **610** coupled with a whipstock assembly **650**. For example, FIG. **6** illustrates the downhole tool **600** as it might appear during it's run-in-hole configuration, wherein the milling tool **610** is similar in many respects to the milling tool **400** and the whipstock assembly **650** is similar in many respects to the whipstock assembly **500**.

[0054] Turning to FIGS. **7A** and **7B**, illustrated are a cross-sectional view and enlarged cross-sectional view, respectively, of the downhole tool **600** of FIG. **6**. FIGS. **7A** and **7B** also illustrate the downhole tool **600** as it might appear during it's run-in-hole configuration. Accordingly, the lug slot(s) **335** of the milling tool **610** are engaged with the lug(s) **535** of the whipstock assembly **650**, and the sliding sleeve **410** is fixed to the mill body **310** using one or more shear features **710**.

[0055] Further to the embodiment of FIGS. **7A** and **7B**, the milling tool **610** may additionally include a ball seat **720** located along an interior surface thereof. As will be discussed further below, the ball seat **720** is configured to engage with a plugging member (e.g., ball member) to close a fluid passageway through the milling tool **610**. The milling tool **610** of FIGS. **7A** and **7B** may additionally include one or more sliding sleeve pressure ports **730** and related seals **740** associated with the sliding sleeve **410**, for example to employ pressure to shift the sliding sleeve **410** from a first position (e.g., as shown in FIGS. **7A** and **7B**) to a second position (e.g., as shown in FIGS. **9A** through **9C**).

[0056] The milling tool **710** of FIGS. **7A** and **7B** may additionally include a snap ring **750**, for example positioned in the sliding sleeve **410**. The snap ring **750**, in one or more embodiments, is configured to engage with a snap ring groove **760** in the mill body **310**. For example, the snap ring **750** and snap ring groove **760** may engage one another once the sliding sleeve **410** moves to the second position, thus fixing the sliding sleeve **410** in the second position during subsequent downhole operations.

[0057] Turning briefly to FIG. **7C**, illustrated is a partial transparent bottom view of the downhole tool **600** of FIG. **7B**. FIG. **7C** illustrates that when the sliding sleeve **410** is in the first position, the finger portion **420** of the sliding sleeve **410** is engaged within the spacing **325**. Accordingly, without removing the finger portion **420** from the spacing **325**, the lug **535** of the whipstock assembly **650** cannot disengage from the lug slot **335** of the milling tool **610**, or vice versa.

[0058] Turning to FIGS. 8A and 8B, illustrated are a cross-sectional view and enlarged cross-sectional view, respectively, of the downhole tool **600** of FIGS. 7A and 7B after dropping a plugging member **810** (e.g., drop ball) within the downhole tool **600**. In the illustrated embodiment, the plugging member **810** seats upon the ball seat **720**, and thus substantially restricts fluid flow through the downhole tool **600**. In one or more embodiments, a shear feature **820** exists between the mill body **310** and the ball seat **720**, such that the ball seat **720** will only shift downhole after a prescribed amount of pressure (e.g., a pressure sufficient to shift the sliding sleeve **410** from its first position to its second position prior to shearing).

[0059] Turning to FIGS. 9A and 9B, illustrated are a cross-sectional view and enlarged cross-sectional view, respectively, of the downhole tool **600** of FIGS. 8A and 8B after pressuring down upon the plugging member **810**. In the illustrated embodiment, fluid pressure within the mill body **310** exits the one or more sliding sleeve pressure ports **730**, thereby employing the seals **740** to shift the sliding sleeve **410** from the first position (e.g., as shown in FIGS. 6 through 7B) to the illustrated second position. As shown, the snap ring **750** may now engage with the snap ring groove **760**, thereby holding the sliding sleeve **410** in the second position. While not shown in the illustrated views, the finger portion **420** is no longer located in the spacing **325**. Nevertheless, the lug slot(s) **335** and lug(s) **535** remain engaged with one another.

[0060] Turning to FIGS. 10A and 10B, illustrated are a cross-sectional view and enlarged cross-sectional view, respectively, of the downhole tool **600** of FIGS. 9A and 9B after continuing to pressuring down upon the plugging member **810**, thereby shifting the ball seat **720** further downhole to re-establish fluid circulation around the plugging member **810** and through the downhole tool **600**. Again, the shear feature **820** holds the ball seat **720** in the first position, and the fluid pressure must overcome the shear value of this shear feature **820** for the ball seat **720** to slide to the second position. Nevertheless, in at least one embodiment the lug slot(s) **335** and lug(s) **535** still remain engaged with one another.

[0061] Turning to FIGS. 11A and 11B, illustrated are a cross-sectional view and enlarged cross-sectional view, respectively, of the downhole tool **600** of FIGS. 10A and 10B after rotating the milling tool **610** relative to the whipstock assembly **650** to free the lug(s) **535** from the lug slot(s) **335**. Typically, the whipstock assembly **650** is rotationally fixed within the wellbore and the milling tool **610** is rotated to free the lug(s) **535**. In at least one embodiment, the milling tool **610** is also being withdrawn uphole as it is being rotated.

[0062] Turning briefly to FIG. 11C, illustrated is a partial transparent bottom view of the downhole tool **600** of FIG. 11B. FIG. 11C illustrates that the sliding of the sliding sleeve **410** from the first position to the second position removes the finger portion **420** of the sliding sleeve **410** from the spacing **325**. Accordingly, the lug(s) **535** of the whipstock assembly **650** are no longer prevented from disengaging from the lug slot(s) **335** of the milling tool **610** by the finger portion **420**, or vice versa. In fact, in the embodiment of FIG. 11C, the lug slot(s) **335** and the lug(s) **535** have rotated relative to one another, and thus the lug(s) **535** are no longer located within the lug slot(s) **335**.

[0063] Turning to FIG. 12, illustrated is a side view of the downhole tool **600** of FIGS. 11A and 11B after the milling tool **610** has fully disengaged from the whipstock assembly **650**. For example, FIG. 12 illustrates the downhole tool **600** as it might appear just prior to it being used to create an opening in wellbore casing and/or a sidetrack in surrounding formation.

[0064] Turning to FIGS. 13A and 13B, illustrated are a cross-sectional view and enlarged cross-sectional view, respectively, of the downhole tool **600** of FIG. 12.

[0065] Turning briefly to FIG. 13C, illustrated is a partial transparent bottom view of the downhole tool **600** of FIG. 13B. FIG. 13C illustrates that the lug(s) **535** are no longer engaged with the lug slot(s) **335** or the spacing **325** between the blades **330**.

[0066] In one or more embodiments, the process would continue by initiating the milling operation. In one or more embodiments, the milling tool **610** would consume the lug(s) **535** and being to mill a window in the casing and/or a sidetrack in the formation. In at least one embodiment, and outside

diameter (OD) of the shifting sleeve **410** is less than the cutting diameter (d.sub.1) of the larger bit assembly, so it can follow in the path created by the larger bit assembly. Also, the shifting sleeve **410** could be made from a material (e.g., aluminum) that could wear away from rubbing against whipstock/casing/formation. Aspects disclosed herein include: [0067] A. A milling tool, the milling tool including: 1) a mill body; 2) a plurality of blades extending radially outward from the mill body, each of the plurality of blades having a primary surface and first and second side surfaces coupled to opposing sides of the primary surface, adjacent blades of the plurality of blades separated by a spacing; and 3) a lug slot extending partially into one of the plurality of blades from the first side surface exposed by the spacing, the lug slot configured to engage with a lug of a related tool to couple the milling tool with the related tool when rotating in a first direction and to disengage from the lug of the related tool to release the milling tool from the related tool when rotating in a second opposite direction. [0068] B. A whipstock assembly, the whipstock assembly including: 1) a whipstock body having an uphole end and a downhole end, the whipstock body defining a taperface; and 2) a lug coupled proximate the uphole end of the whipstock body, the lug configured to engage with a lug slot of a related milling tool to couple the whipstock assembly with the related milling tool when the milling tool rotates in a first direction and to disengage from the lug slot of the related milling tool to release the whipstock assembly from the related milling tool when the milling tool rotates in a second opposite direction. [0069] C. A well system, the well system including: 1) a wellbore located through one or more subterranean formations; and 2) a downhole tool located within the wellbore, the downhole tool including: a) a whipstock assembly, the whipstock assembly including: i) a whipstock body having an uphole end and a downhole end, the whipstock body defining a taperface; and ii) a lug coupled proximate the uphole end of the whipstock body; and b) a milling tool, the milling tool including: i) a mill body; ii) a plurality of blades extending radially outward from the mill body, each of the plurality of blades having a primary surface and first and second side surfaces coupled to opposing sides of the primary surface, adjacent blades of the plurality of blades separated by a spacing; and iii) a lug slot extending partially into one of the plurality of blades from the first side surface exposed by the spacing, wherein the lug slot is configured to engage with the lug of the whipstock assembly to couple the milling tool with the whipstock assembly when rotating in a first direction and to disengage from the lug of the whipstock assembly to release the milling tool from the whipstock assembly when rotating in a second opposite direction. [0070] D. A method, the method including: 1) running a downhole within a wellbore, the downhole tool including: a) a whipstock assembly, the whipstock assembly including: i) a whipstock body having an uphole end and a downhole end, the whipstock body defining a taperface; and ii) a lug coupled proximate the uphole end of the whipstock body; and b) a milling tool coupled to the whipstock assembly, the milling tool including: i) a mill body; ii) a plurality of blades extending radially outward from the mill body, each of the plurality of blades having a primary surface and first and second side surfaces coupled to opposing sides of the primary surface, adjacent blades of the plurality of blades separated by a spacing; and iii) a lug slot extending partially into one of the plurality of blades from the first side surface exposed by the spacing, wherein the lug slot is configured to engage with the lug of the whipstock assembly to couple the milling tool with the whipstock assembly when rotating in a first direction and to disengage from the lug of the whipstock assembly to release the milling tool from the whipstock assembly when rotating in a second opposite direction; and 2) rotating the downhole tool in the second direction to disengage the milling tool from the whipstock assembly.

Aspects A, B, C and D may have one or more of the following additional elements in combination: Element 1: wherein the lug slot has a larger lug slot portion radially proximate the mill body and a smaller lug slot portion radially distal the mill body. Element 2: wherein the lug slot is an inverted T-shaped lug slot. Element 3: wherein the lug slot is a first lug slot, and further including a second lug slot extending partially into one of the plurality of blades from the first side surface exposed by the spacing. Element 4: wherein the first lug slot and the second lug slot are located in a same one

of the plurality of blades. Element 5: wherein the first lug slot is located in a first of the plurality of blades and the second lug slot is located in a second of the plurality of blades. Element 6: further including a sliding sleeve having a finger portion, the sliding sleeve positioned about the mill body proximate the plurality of blades, the sliding sleeve configured to move between a first position locating the finger portion within the spacing proximate the lug slot and a second position removing the finger portion from the spacing proximate the lug slot, the first position configured to keep the lug of the related tool engaged within the lug slot and the second position configured to allow the lug of the related tool to disengage from the lug slot. Element 7: wherein the plurality of blades are angled relative to an axis of rotation of the mill body, and further wherein the finger portion is angled relative to the axis of rotation of the mill body. Element 8: wherein a first angle of the plurality of blades is substantially identical to a second angle of the finger portion. Element 9: wherein the sliding sleeve is a hydraulically activated sliding sleeve, an increase in pressure within the mill body configured to move the sliding sleeve from the first position to the second position. Element 10: wherein the lug has a smaller lug portion and a larger lug portion, the smaller lug portion more proximate the whipstock body than the larger lug portion. Element 11: wherein the lug is a T-shaped lug, the smaller lug portion coupled to the whipstock body. Element 12: wherein the lug is a first lug, and further including a second lug coupled proximate the uphole end of the whipstock body. Element 13: wherein a centerline taken through the first lug and the second lug is angled relative to a longitudinal centerline of the whipstock body. Element 14: wherein an angle ($\theta_{sub.3}$) between the centerline taken through the first lug and the second lug and the longitudinal centerline is substantially identical to an angle ($\theta_{sub.1}$) of a plurality of blades of the related milling tool that the whipstock assembly is configured to engage. Element 15: wherein the lug is integrally formed with the whipstock body. Element 16: wherein the lug is coupled to the whipstock body via an opening in the whipstock body. Element 17: wherein a fastener extends at least partially into the opening to couple the lug to the whipstock body. Element 18: wherein the lug is coupled to a concave portion of the whipstock body. Element 19: wherein the lug slot has a larger lug slot portion radially proximate the mill body and a smaller lug slot portion radially distal the mill body, and the lug has a smaller lug portion and a larger lug portion, the smaller lug portion more proximate the whipstock body than the larger lug portion. Element 20: wherein the lug slot is an inverted T-shaped lug slot and the lug is a T-shaped lug, the smaller lug portion coupled to the whipstock body. Element 21: wherein the lug slot is a first lug slot, and further including a second lug slot extending partially into one of the plurality of blades from the first side surface exposed by the spacing, and further wherein the lug is a first lug, and further including a second lug coupled proximate the uphole end of the whipstock body. Element 22: wherein the first lug slot and the second lug slot are located in a same one of the plurality of blades. Element 23: wherein the milling tool further includes a sliding sleeve having a finger portion, the sliding sleeve positioned about the mill body proximate the plurality of blades. Element 24: wherein the sliding sleeve is configured to move between a first position locating the finger portion within the spacing proximate the lug slot and a second position removing the finger portion from the spacing proximate the lug slot, the first position configured to keep the lug of the whipstock assembly engaged within the lug slot and the second position configured to allow the lug of the whipstock assembly to disengage from the lug slot. Element 25: wherein the lug slot is engaged with the lug, and further wherein the sliding sleeve is in the first position keeping the lug of the whipstock assembly engaged within the lug slot. Element 26: wherein the lug slot is engaged with the lug, and further wherein the sliding sleeve is in the second position allowing the lug of the whipstock assembly to disengage from the lug slot. Element 27: wherein the sliding sleeve is a hydraulically activated sliding sleeve, an increase in pressure within the mill body configured to move the sliding sleeve from the first position to the second position. Element 28: wherein the milling tool further includes a sliding sleeve having a finger portion, the sliding sleeve positioned about the mill body proximate the plurality of blades. Element 29: wherein the sliding sleeve is configured to move between a first position locating the

finger portion within the spacing proximate the lug slot and a second position removing the finger portion from the spacing proximate the lug slot, the first position configured to keep the lug of the whipstock assembly engaged within the lug slot and the second position configured to allow the lug of the whipstock assembly to disengage from the lug slot. Element 30: wherein running the downhole tool within the wellbore include running the downhole tool within the wellbore with the sliding sleeve in the first position. Element 31: further including moving the sliding sleeve from the first position to the second position after running the downhole tool within the wellbore and before rotating the downhole tool in the second direction.

[0071] Those skilled in the art to which this application relates will appreciate that other and further additions, deletions, substitutions and modifications may be made to the described embodiments.

Claims

1. A milling tool, comprising: a mill body; a plurality of blades extending radially outward from the mill body, each of the plurality of blades having a primary surface and first and second side surfaces coupled to opposing sides of the primary surface, adjacent blades of the plurality of blades separated by a spacing; and a lug slot extending partially into one of the plurality of blades from the first side surface exposed by the spacing, the lug slot configured to engage with a lug of a related tool to couple the milling tool with the related tool when rotating in a first direction and to disengage from the lug of the related tool to release the milling tool from the related tool when rotating in a second opposite direction.
 2. The milling tool as recited in claim 1, wherein the lug slot has a larger lug slot portion radially proximate the mill body and a smaller lug slot portion radially distal the mill body.
 3. The milling tool as recited in claim 2, wherein the lug slot is an inverted T-shaped lug slot.
 4. The milling tool as recited in claim 1, wherein the lug slot is a first lug slot, and further including a second lug slot extending partially into one of the plurality of blades from the first side surface exposed by the spacing.
 5. The milling tool as recited in claim 4, wherein the first lug slot and the second lug slot are located in a same one of the plurality of blades.
 6. The milling tool as recited in claim 4, wherein the first lug slot is located in a first of the plurality of blades and the second lug slot is located in a second of the plurality of blades.
 7. The milling tool as recited in claim 1, further including a sliding sleeve having a finger portion, the sliding sleeve positioned about the mill body proximate the plurality of blades, the sliding sleeve configured to move between a first position locating the finger portion within the spacing proximate the lug slot and a second position removing the finger portion from the spacing proximate the lug slot, the first position configured to keep the lug of the related tool engaged within the lug slot and the second position configured to allow the lug of the related tool to disengage from the lug slot.
 8. The milling tool as recited in claim 7, wherein the plurality of blades are angled relative to an axis of rotation of the mill body, and further wherein the finger portion is angled relative to the axis of rotation of the mill body.
 9. The milling tool as recited in claim 8, wherein a first angle of the plurality of blades is substantially identical to a second angle of the finger portion.
 10. The milling tool as recited in claim 7, wherein the sliding sleeve is a hydraulically activated sliding sleeve, an increase in pressure within the mill body configured to move the sliding sleeve from the first position to the second position.
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